

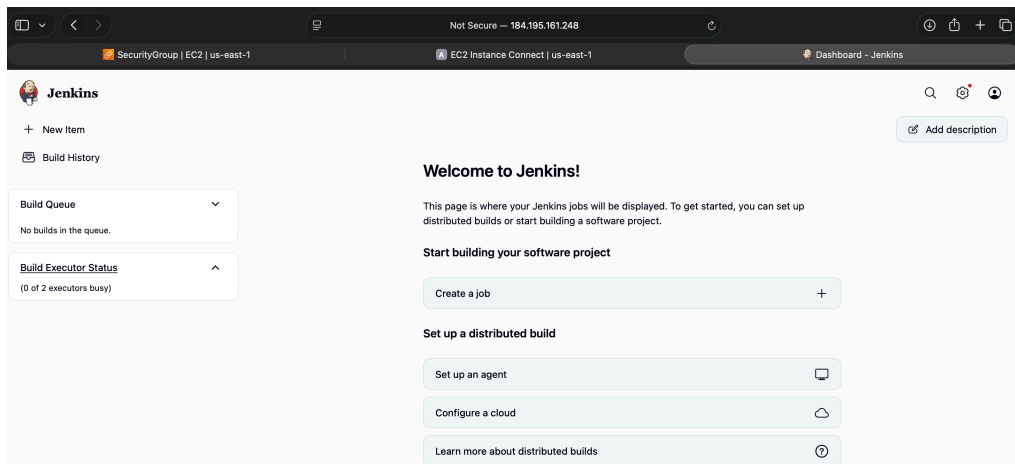
Jenkins-Task-1

1. Install Jenkins and run Jenkins on port number 8081.

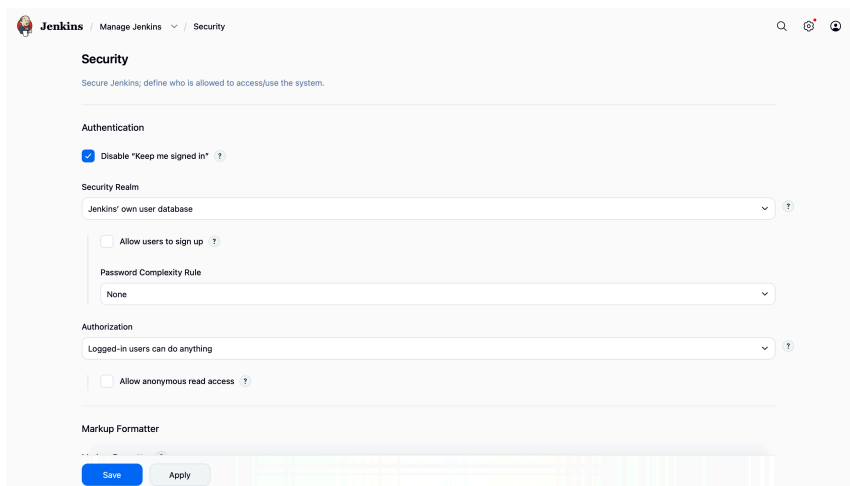
```
Installed:
jenkins-2.50.1-1.noarch

Complete!
[ec2-user@ip-10-0-7-53 ~]$ sudo systemctl enable jenkins
sudo systemctl start jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[ec2-user@ip-10-0-7-53 ~]$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-30 16:50:14 UTC; 17s ago
     Main PID: 27132 (java)
       Tasks: 42 (limit: 1059)
      Memory: 955.8M
         CPU: 19.776s
    CGroup: /system.slice/jenkins.service
            └─27132 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

[ec2-user@ip-10-0-7-53 ~]$ sudo systemctl restart jenkins
[ec2-user@ip-10-0-7-53 ~]$ sudo ss -ltnp | grep 8081
LISTEN 0      50          *:8081      *:8081      users: (("java",pid=27983,fd=9))
[ec2-user@ip-10-0-7-53 ~]$
```

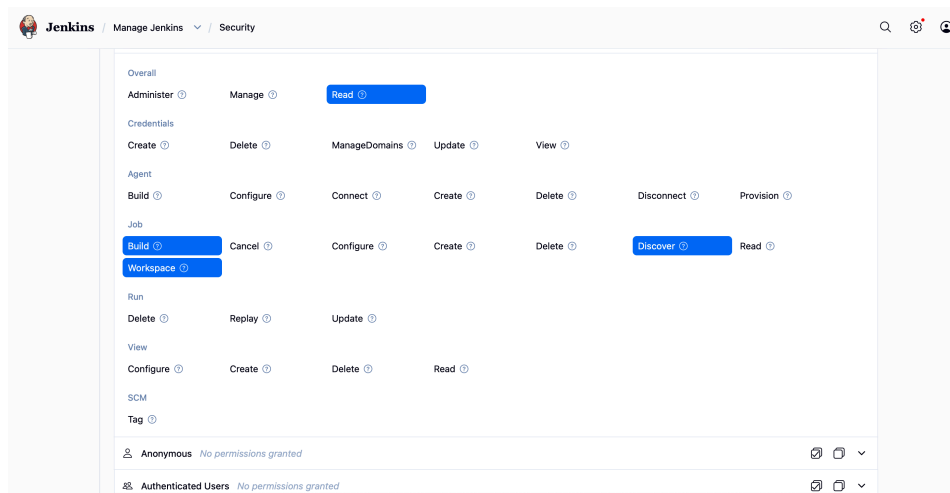
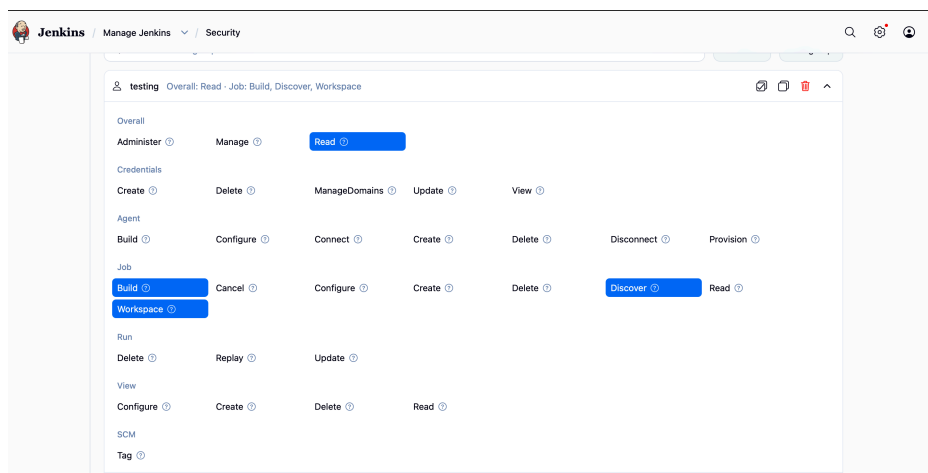
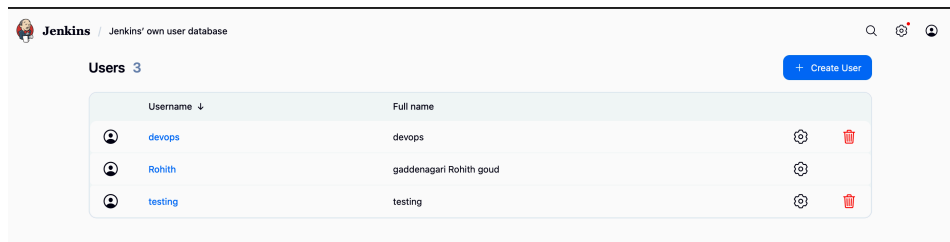


2. Secure Jenkins server



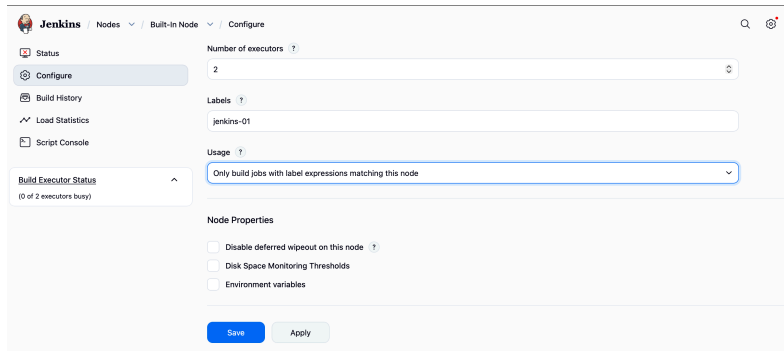
Jenkins-Task-1

3. Create users called DevOps, Testing in Jenkins with Limited access.

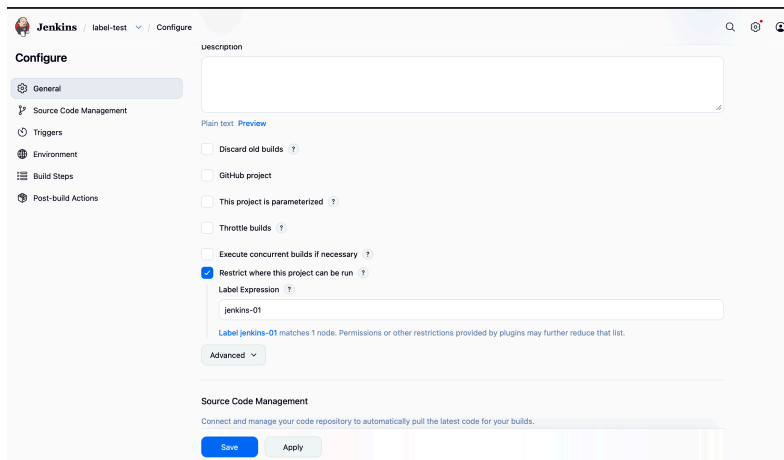


Jenkins-Task-1

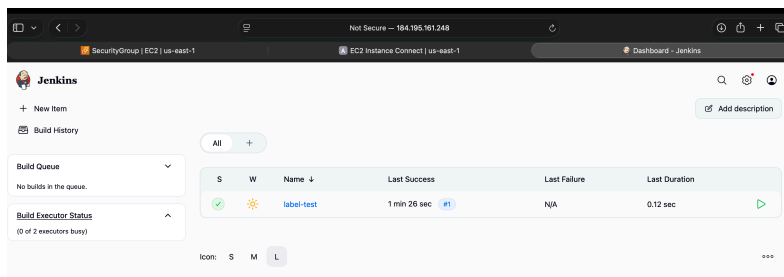
4. Configure labels and restrict the jobs to execute based on label only.



This screenshot shows the Jenkins configuration page for a specific node. The left sidebar contains links for Status, Configure, Build History, Load Statistics, and Script Console. The main area is titled 'Configure' and includes a 'Build Executor Status' section showing 0 of 2 executors busy. The 'Number of executors' is set to 2. The 'Labels' field contains 'jenkins-01'. The 'Usage' dropdown is set to 'Only build jobs with label expressions matching this node'. Under 'Node Properties', there are checkboxes for 'Disable deferred wipeout on this node', 'Disk Space Monitoring Thresholds', and 'Environment variables', all of which are currently unchecked. 'Save' and 'Apply' buttons are at the bottom.



This screenshot shows the configuration page for a Jenkins job named 'label-test'. The left sidebar includes links for General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is active, showing a 'Description' field, a 'Plain text' preview, and several checkboxes: 'Discard old builds', 'GitHub project', 'This project is parameterized', 'Throttle builds', 'Execute concurrent builds if necessary', and 'Restrict where this project can be run'. The 'Restrict where this project can be run' checkbox is checked, and the 'Label Expression' field below it contains 'jenkins-01'. A message states 'Label jenkins-01 matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.' Below this is an 'Advanced' dropdown. The 'Source Code Management' section at the bottom has a 'Save' button and an 'Apply' button.



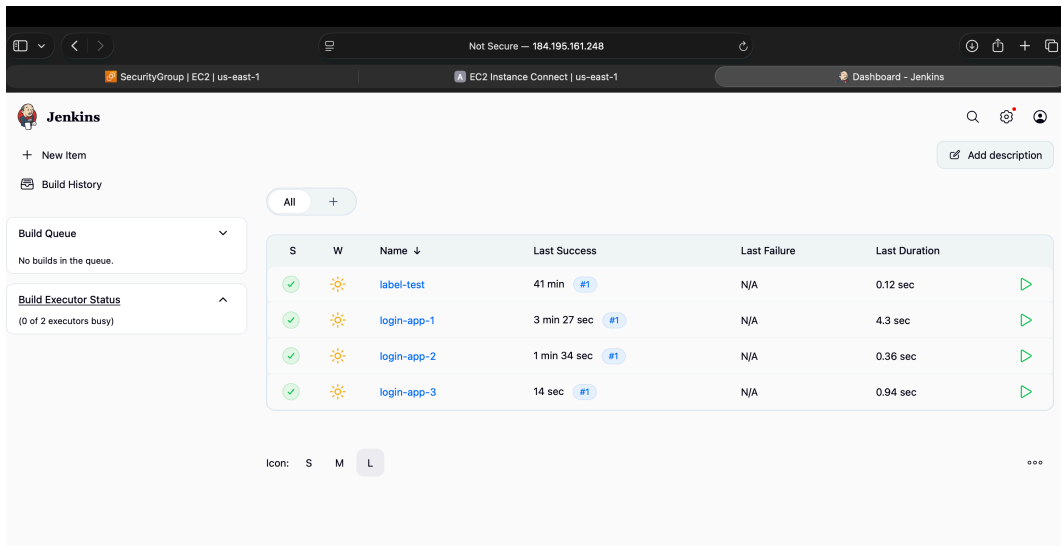
This screenshot shows the Jenkins dashboard. The top bar indicates the browser is 'Not Secure' and the URL is '184.195.161.248'. The dashboard includes a 'New Item' button, a 'Build History' link, and a 'Build Queue' section showing 'No builds in the queue.' The 'Build Executor Status' section shows '0 of 2 executors busy'. The main table displays the build history for the 'label-test' job.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	label-test	1 min 26 sec #1	N/A	0.12 sec

At the bottom, there is a 'Icon' section with buttons for 'S', 'M', and 'L'.

Jenkins-Task-1

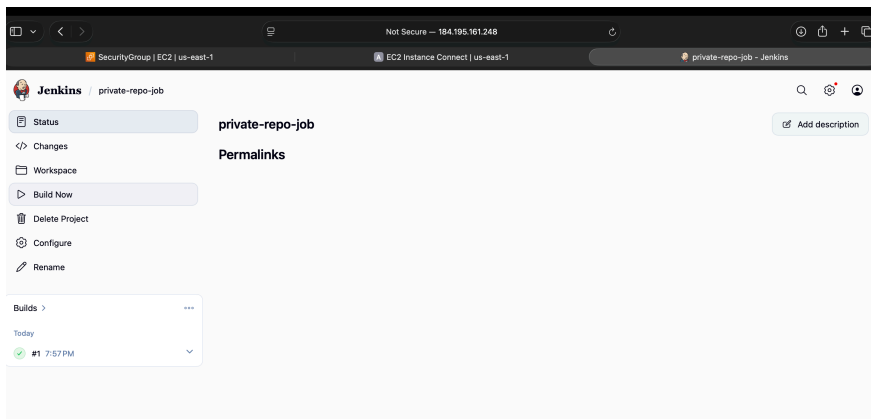
5. Create Three sample jobs using the below URL.
 - **Source Code:** https://github.com/betawins/Techie_horizon_Login_app.git



The screenshot shows the Jenkins Dashboard. On the left, there's a sidebar with 'New Item', 'Build History', 'Build Queue' (showing 'No builds in the queue.'), and 'Build Executor Status' (showing '0 of 2 executors busy'). The main area displays a table of jobs with columns: S, W, Name, Last Success, Last Failure, and Last Duration. The jobs listed are 'label-test', 'login-app-1', 'login-app-2', and 'login-app-3'. All jobs show a green status icon and a sun icon. The 'Last Success' column shows the time since the last successful build, and the 'Last Duration' column shows the build duration.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	label-test	41 min #1	N/A	0.12 sec
✓	☀	login-app-1	3 min 27 sec #1	N/A	4.3 sec
✓	☀	login-app-2	1 min 34 sec #1	N/A	0.36 sec
✓	☀	login-app-3	14 sec #1	N/A	0.94 sec

6. Create one Jenkins job using GitHub Private repository.



The screenshot shows the Jenkins job configuration page for 'private-repo-job'. The left sidebar contains links for 'Status', 'Changes', 'Workspace', 'Build Now', 'Delete Project', 'Configure', and 'Rename'. The main area shows the job name 'private-repo-job' and a 'Permalinks' section. Below the job name, there's a 'Builds' section showing a list of builds, with the first build '#1' at '7:57 PM' marked as successful with a green checkmark.